

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Display
Manufacturer: BenQ
Firmware Version: N/A
Model(s): ST4301K, ST5501K, ST6501K

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.03.0000-b006	2.3.0

Version History

Module Version	Date	Notes
1_0_0_0	6/26/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '98', and 'Broadcast'. If Device ID is set to 'Broadcast', status is unavailable.
Example: `InterfaceName.DeviceID = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='ST4301K')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='ST4301K')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4660, Model='ST4301K')</code>

Control Commands

Format with Qualifier:

InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})

Format with Qualifier:

InterfaceName.Set(Command, Value)

Command AspectRatio	Value 'Full' '16:9'	Value '4:3'	Value '1:1'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Full')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
Qualifier Key 'Type'	Qualifier Value 'Button'	Qualifier Value 'IR'	Qualifier Value 'Button and IR'
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On', {'Type': 'Button'})			
Command Input	Value 'VGA' 'DVI-D'	Value 'HDMI 1' 'DisplayPort'	Value 'HDMI 2' 'Android'
# Input example InterfaceName.Set('Input', 'VGA')			
Command MenuNavigation	Value 'Up' 'Right' 'Exit'	Value 'Down' 'Ok'	Value 'Left' 'Menu'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command PictureMode	Value 'Standard' 'Custom'	Value 'Bright'	Value 'Soft'
# PictureMode example InterfaceName.Set('PictureMode', 'Standard')			
Command Power	Value 'On'	Value 'Off'	Value 'Android Off'
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Full' '16:9'	Value '4:3'	Value '1:1'
InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
Qualifier Key 'Type'	Qualifier Value 'Button'	Qualifier Value 'IR'	Qualifier Value 'Button and IR'
InterfaceName.Update('ExecutiveMode', {'Type': 'Button'}) Value = InterfaceName.ReadStatus('ExecutiveMode', {'Type': 'Button'}) InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command Input	Value 'VGA' 'DVI-D'	Value 'HDMI 1' 'DisplayPort'	Value 'HDMI 2' 'Android'
InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command PictureMode	Value 'Standard' 'Custom'	Value 'Bright'	Value 'Soft'
InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	Value 'Android Off'
InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power')			

Global Scripter Module Communication Sheet

InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)	
Command	Value
Volume	0 to 100 in steps of 1
InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)	

Cable and Adapter Requirements

Captive Screw to 3.5mm TRS Connector

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

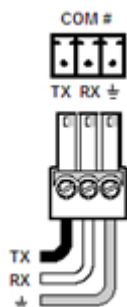
Data Bits: 8

Parity: None

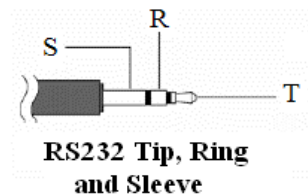
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Please contact the manufacturer for information regarding the pin assignments diagram.



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type:	Ethernet
Default Port:	4660
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 1:1	801s1002\x0D
Aspect Ratio 16:9	801s1003\x0D
Aspect Ratio 4:3	801s1001\x0D
Aspect Ratio Full	801s1000\x0D
Audio Mute Off	801s6000\x0D
Audio Mute On	801s6001\x0D
Auto Image None	801s\x8F\x0D
Executive Mode Off Type Button	801sE001\x0D
Executive Mode Off Type Button and IR	801sC001\x0D
Executive Mode Off Type IR	801sB001\x0D
Executive Mode On Type Button	801sE000\x0D
Executive Mode On Type Button and IR	801sC000\x0D
Executive Mode On Type IR	801sB000\x0D
Input Android	801s"101\x0D
Input DisplayPort	801s"007\x0D
Input DVI-D	801s"006\x0D
Input HDMI 1	801s"001\x0D
Input HDMI 2	801s"002\x0D
Input VGA	801s"000\x0D
Menu Navigation Down	801s@011\x0D
Menu Navigation Exit	801s@022\x0D
Menu Navigation Left	801s@012\x0D
Menu Navigation Menu	801s@020\x0D
Menu Navigation Ok	801s@014\x0D
Menu Navigation Right	801s@013\x0D
Menu Navigation Up	801s@010\x0D
Picture Mode Bright	801s\x81001\x0D
Picture Mode Custom	801s\x81003\x0D
Picture Mode Soft	801s\x81002\x0D
Picture Mode Standard	801s\x81000\x0D
Power Android Off	801s!002\x0D
Power Off	801s!000\x0D
Power On	801s!001\x0D
Volume 0	801s5000\x0D
Volume 100	801s5100\x0D

Appendix B. Update Commands

Aspect Ratio	801gw000\x0D
Audio Mute	801gg000\x0D
Executive Mode Type Button	801gs000\x0D
Executive Mode Type Button and IR	801gi000\x0D
Executive Mode Type IR	801gh000\x0D
Input	801gj000\x0D
Picture Mode	801g\xB1000\x0D
Power	801gl000\x0D
Volume	801gf000\x0D